

Scientific Computation

MECHTRON 3X03
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Lecture 9 - Partial Pivoting

Total Complexity of Gaussian Elimination + Backward Substitution

Let's finish up the complexity analysis from last class:

Gaussian Elimination: $\frac{2}{3}n^3 - \frac{1}{2}n^2 - \frac{1}{6}n$

Backwards substitution: n^2

$$\text{Total: } \frac{2}{3}n^3 + \frac{1}{2}n^2 - \frac{1}{6}n = \frac{2}{3}n^3 + O(n^2)$$

So the overall complexity is $O(n^3)$

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Lecture Outline

- **Motivating example**
- Gaussian elimination with partial pivoting (GEPP)

A Motivating Example

A Motivating Example

Consider the system

$$\begin{aligned} 10^{-5}x_1 + x_2 &= 1 \\ 2x_1 + x_2 &= 2. \end{aligned}$$

$$A = \begin{bmatrix} 10^{-5} & 1 \\ 2 & 1 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

The solution can be computed in Julia (with double precision):

```
A = [1e-5 1
      2    1]
b = [1.; 2.]
x = A\b

✓ 0.0s

2-element Vector{Float64}:
 0.5000025000125
 0.999994999975
```

What happens if we try Gaussian elimination in a lower precision FP system?

Naive Gaussian Elimination

Example 1: Let's solve the same system with a 5-digit decimal floating point representation (i.e., $t = 5$ and $\beta = 10$):

$$\begin{aligned} 10^{-5}x_1 + x_2 &= 1 \\ 2x_1 + x_2 &= 2 \end{aligned} \iff \begin{bmatrix} 10^{-5} & 1 \\ 2 & 1 \end{bmatrix} x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Step 1: eliminate 2 with the first row, also called the pivot row $k=1$

on step k of GE, the k^{th} element of pivot row ($\text{row } k$) is called the **pivot** = 10^{-5} for this step.

$$2x_1 + x_2 - 2 - \left(\frac{2}{10^{-5}}\right) [10^{-5}x_1 + x_2 - 1] = 0$$

$$(1 - 2 \cdot 10^5)x_2 = 2 - 2 \cdot 10^5$$

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An Inaccurate Result

Example 1 cont.: With our approximate solution $\tilde{x}_2$, solve for $\tilde{x}_1$:

step 2: Backward substitution:

Recall $r=5$. so $fl(1 - 2 \times 10^5) = -2 \times 10^5 \therefore -2 \times 10^5 x_2 = -2 \times 10^5$
 $fl(2 - 2 \times 10^5) = -2 \times 10^5$ and $\tilde{x}_2 = 1.00000$

$$10^{-5}x_1 + x_2 = 1 \Rightarrow 10^{-5}\tilde{x}_1 + \tilde{x}_2 = 1 \therefore \tilde{x}_1 = \frac{1 - 1.00000}{10^{-5}} = 0$$

But: true solution $x_1^* = 0.5$

The abs. error in $\tilde{x}_2$ was: $\tilde{x}_2 - x_2^* = 1 - 9.99994 \times 10^{-1} \approx 5 \times 10^{-6}$

The abs. error in $\tilde{x}_1$ is 0.5.

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Error Analysis

Example 1 cont.: Let's look at the source of the error in $\tilde{x}_1$:

$$\tilde{x}_1 = \frac{1 - \tilde{x}_2}{10^{-5}} \approx \frac{1 - (x_2^* + \overbrace{5 \cdot 10^{-6}}^{\text{abs. error}})}{10^{-5}}$$

$\therefore \approx 0.5 - 1 = 10^{-5} \cdot 9.99994 = (FP64) = 10^{-6}$

0.5, which is 10⁻⁵ times the pivot

$$\approx \frac{1 - x_2^*}{10^{-5}} = 5 \times 10^{-6} \cdot \frac{1}{10^{-5}}$$

$$= x_1^* - \frac{\delta_2}{\text{pivot}}$$

$$= x_1^* - 0.5 = x_1^* - \delta_1$$

$$\delta_1 = \delta_2 \cdot \frac{1}{\text{pivot}}$$

The error in $\tilde{x}_2$ is multiplied by $\frac{1}{\text{pivot}} = 10^5$

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Solution: Avoid Small Pivots

Example 2: Let's solve the same system but swap the rows:

$$\begin{aligned} 2x_1 + x_2 &= 2 \\ 10^{-5}x_1 + x_2 &= 1 \end{aligned} \iff \begin{bmatrix} 2 & 1 \\ 10^{-5} & 1 \end{bmatrix} x = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

multiply the first row by $\frac{10^{-5}}{2}$:

$$10^{-5}x_1 + \frac{10^{-5}}{2}x_2 = \frac{10^{-5}}{2} \cdot 2$$

and subtract from the second row:

$$\left(1 - \frac{10^{-5}}{2}\right)x_2 = 1 - 10^{-5}$$

$$\text{but } \beta = 10, t = 5 \rightarrow 1 - \frac{10^{-5}}{2} \text{ and } 1 - 10^{-5} = 1.0000$$

$$1 \cdot \tilde{x}_2 = 1, \text{ so } \tilde{x}_2 = 1$$

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Solution: Avoid Small Pivots

Example 2 cont.: We now have the system in row echelon form:

$$A|b \leftarrow \left[\begin{array}{cc|c} 2 & 1 & 2 \\ 0 & 1 & 1 \end{array} \right]$$

Writing out the first equation:

$$2\tilde{x}_1 + \tilde{x}_2 = 2$$

$$\tilde{x}_1 = \frac{2 - \tilde{x}_2}{2}$$

$$= 0.5$$

now, $\tilde{x}_2 \approx x_2^* + 5 \times 10^{-6}$, so:

$$\tilde{x}_1 = \frac{2 - \tilde{x}_2}{2} \approx \frac{2 - (x_2^* + 5 \times 10^{-6})}{2} = \frac{2 - x_2^*}{2} - 5 \times 10^{-6}$$

abs error in $\tilde{x}_1$
 $\uparrow$
 δ_1

$$\text{So, error in } \tilde{x}_1: \delta_1 = \frac{\delta_2}{\text{pivot}} \approx -2.5 \times 10^{-6}$$

x_1^*

Lecture Outline

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Gaussian Elimination with Partial Pivoting (GEPP)



Straightforward idea: at step k , eliminate with the row that has the largest absolute value entry in the k^{th} column

I.e., maximize the **pivot**: $\begin{bmatrix} 10^{-5} & 1 \\ 2 & 1 \end{bmatrix} x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

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GEPP In Action

Example 3: Let's solve a slightly larger ($n = 3$) system, still with $t = 5$ and $\beta = 10$:

$$\begin{aligned} x_1 + x_2 + x_3 &= 1 \\ x_1 + 1.0001x_2 + 2x_3 &= 2 \end{aligned} \iff \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1.0001 & 2 \end{bmatrix} x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

↗ break tie by choosing 1st row

$$x_1 + 2x_2 + 2x_3 = 3 \quad [1 \quad 2 \quad 2] \quad [3]$$

First row as pivot row: $x_1 + x_2 + x_3 = 1$
 $\rightarrow 0.0001x_2 + x_3 = 1$
 $x_2 + x_3 = 2$

Eliminate with 3rd row as pivot row

$$x_1 + x_2 + x_3 = 1$$

$$0.0001x_2 + x_3 - \left(\frac{0.0001}{1}\right)(x_2 + x_3) = 1 - 2 \cdot \frac{0.0001}{1}$$

$$\rightarrow (1 - 0.0001)x_3 = 1 - 0.0002$$

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GEPP In Action

Example 3 cont.: Let's check the solution in Julia with $x = A \setminus b$ again for

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1.0001 & 2 \\ 1 & 2 & 2 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix},$$

which gives us $x = [-1, 1.000100010001, 9.99899989999 \cdot 10^{-1}]^T$

In 5 digit arithmetic,

$$0.9999 \tilde{x}_3 = 0.9998$$

$$\tilde{\tilde{x}}_3 = 9.9990 \times 10^{-1}$$

$$\rightarrow \text{error} \leq 10^{-8}$$

$$\tilde{\tilde{x}}_1 = 1 - \tilde{\tilde{x}}_2 - \tilde{\tilde{x}}_3 = -1 \rightarrow \text{error} \leq 0$$

so, abs. errors are all very small

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GEPP In Action

Example 3 cont.: If we had naively applied GE without pivoting, the second step would be

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0.0001 & 1 \\ 0 & 1 & 1 \end{bmatrix} x = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

$$x_2 + x_3 - 2 - \left(\frac{1}{0.0001}\right) [0.0001 \cdot x_2 + x_3 - 1] = 0$$

$$(1 - 10^4) x_3 = 2 - 10^4$$

$$\therefore 9.9990 \times 10^3 x_3 = -9.9980 \times 10^3$$

$$\therefore \tilde{x}_3 = 9.9990 \times 10^{-1}$$

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GEPP In Action

Example 3 cont.: Finally, let's compare the absolute values of the errors with and without pivoting. We have:

$$\text{Error in } \tilde{x}_3 \approx 10^{-8} \quad \uparrow \text{abs error}$$

$$\tilde{x}_1 = \frac{1 - \tilde{x}_3}{0.0001} = (1 - 0.9990 - \delta_3) \cdot 10^4$$

$$\approx 10^4 \text{ larger than before}$$

在科学计算中，我们不用把matrix化简成RREF，而是直接化成REF就行，然后直接用FP operation带入运算看结果。

为什么会出现cancellation:

首先，由于消元法的特性（注意，这里我们不是按照1ZC3那样，先把当先row的pivot消成1，所以事实上需要用一个比率m乘上pivot row，然后让下面逐行减去），pivot太小会导致即使后面row求解出来的variable都在可接受的误差区间，但是pivot在求解pivot行的未知数x的时候，pivot参与到分母的计算，这让误差被放大很多倍）。

同时，如果精度有限（如果t很小），那么m由于pivot很小会导致m变得很大，所以在后续的运算中，精度会被截断，出现类似cancellation的情况。